

A Counterfactual & Decision-Aware AI Framework for Cost-Optimized Industrial PM & FFS

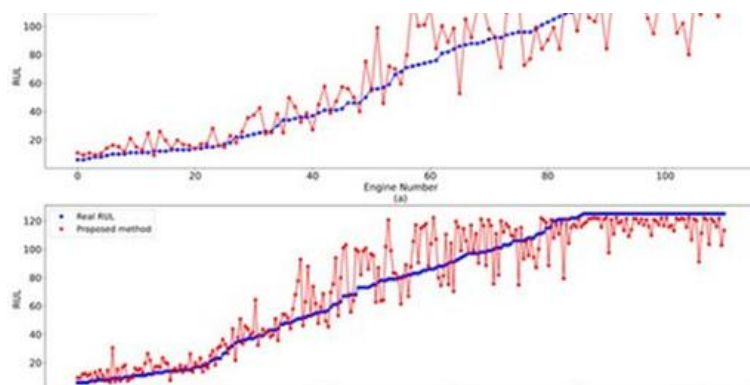
MOTIVATION

Industrial assets like turbofan engines endure punishing mechanical and thermal stresses where sudden failures trigger economic losses, safety hazards and mission-critical downtime. Reactive maintenance responds post-damage; calendar-based schedules over-maintain healthy assets neither tracks real-time health. Predictive models forecast failures but ignore the key question: under economic uncertainty, what's the cost-optimal action? This gap demands an AI framework that predicts degradation, prescribes optimal maintenance, and simulates "what-if" scenarios for strategic decisions.

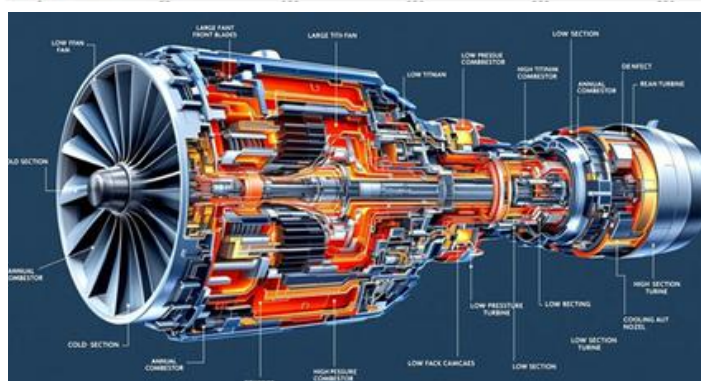
DATASET DETAILS

The dataset utilized in this work is the NASA CMAPSS Turbofan Engine Dataset, which consists of multivariate time-series data from multiple turbofan engines run to failure. The dataset comprises sensor readings 21 sensor channels capturing fan speed, core speed, temperatures, pressures and operating conditions (altitude, Mach number, throttle) per engine cycle capturing full degradation trajectories from healthy state to failure across four fault modes.

Sample RUL Trajectories across CMAPSS subsets (FD001-FD004)

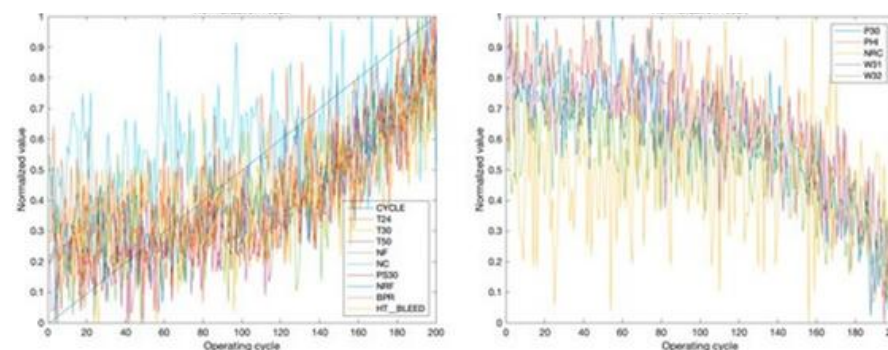


Turbofan engine cross-section highlighting sensor measurement locations

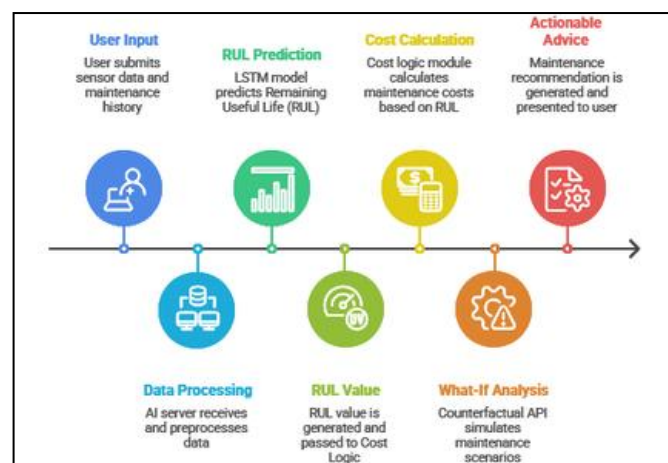


METHODOLOGY

The proposed system is a six-layer decision intelligence framework for turbofan predictive maintenance where Data Ingestion and Preprocessing cleans, normalizes (z-score) and segments multivariate sensor data, followed by Health and Degradation Modeling to generate a normalized Health Index (HI) from 1 (healthy) to 0 (failed). An LSTM-based Failure Forecasting System (FFS) captures temporal patterns detects early degradation and updates predictions continuously. These outputs enable Decision-Aware Cost Optimization, integrating preventive, failure and delay costs (C_m , C_f , C_d) to select optimal actions such as repair, deferment, or run-to-failure. A Counterfactual Reasoning Engine evaluates "what-if" strategies, while Continuous Learning and Feedback improves accuracy and long-term maintenance decisions using real operational data.



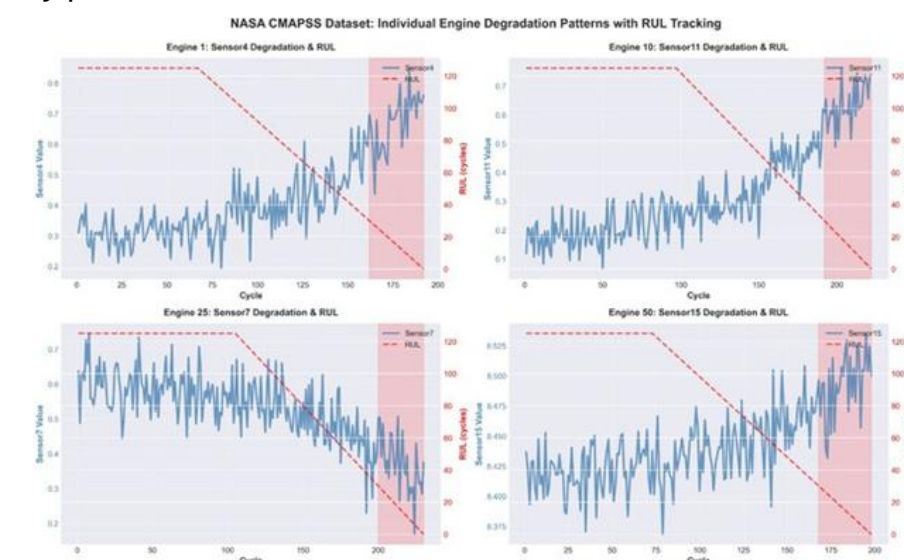
Normalized multivariate sensor trajectories used for Degradation Modeling



Block Diagram depicting System Workflow of PM & FFS

EXPERIMENTAL RESULTS

Reactive maintenance operates on a "wait-until-failure" model, which results in emergency repairs, rushed spare parts and extended downtime making it 3–5× more expensive than planned maintenance. Predictive maintenance overcomes this by using AI/ML models such as LSTM based forecasting to continuously track equipment health, identify early signs of degradation and predict failures in advance. This allows maintenance to be scheduled proactively reducing costs by 20–50% while improving system reliability and uptime. At the decision level, cost-aware optimization compares possible actions repair now, delay maintenance or continue monitoring using expected total cost (ETC) that incorporates failure probability and future consequences. Counterfactual "what-if" analysis further explains the impact of each choice, helping justify proactive actions.



Individual engine degradation patterns with RUL tracking

DETAILS:

Name: TUSHAR MISHRA, KUSHAL SINGH, ARCHIT YADAV, MAYANK SINGH

Enrollment No.: A2305224444, 411, 443, 425

Department: COMPUTER SCIENCE & ENGINEERING

Institute: ASET, AUUP

Email ID: tusharmishra0710@gmail.com